



Dumindu Kumarapeli

DATA SCIENCE UNDERGRADUATE · ML / RESEARCH ENGINEER

I build research prototypes that ship — full pipelines from data generation through deployment, across retrieval-augmented & agentic LLM systems, applied ML and statistics, and computer vision. Most of it runs on a 6 GB consumer GPU.

- ✉ duminduku.25@gmail.com · 🌐 Sri Lanka
- 🔗 linkedin.com/in/dumindu-kumarapeli

EDUCATION

BSc (Hons) Data Science

NSBM Green University, Sri Lanka
UNDERGRADUATE · IN PROGRESS

LEADERSHIP

Assistant Secretary

NSBM AI Association (NAIA)

TECHNICAL SKILLS

LANGUAGES

Python · Java · JavaScript · TypeScript · R

ML / DATA SCIENCE

PyTorch · TensorFlow · scikit-learn · Pandas · Polars · NumPy · SciPy · YOLOv8 · OpenCV

LLM / RAG

Gemini · ChromaDB · Ollama · MCP

BACKEND / INFRA

FastAPI · Spring Boot · PostgreSQL · Docker · Git

FRONTEND / VIZ

React · Next.js · Node.js · Streamlit · Plotly · Power BI

FOCUS & INTERESTS

Medical AI · Epidemiological forecasting · Reinforcement learning

Currently exploring CUDA acceleration, sequence models (RNN/LSTM), and production LLM/RAG pipelines.

- ✉ kadpkumarapeli.workers.dev 🔗 github.com/Dumee-25

01 SELECTED SYSTEMS

OWL — Datacenter AI Operations Platform

An agentic, RAG-powered operations assistant for datacenter infrastructure, with real-time anomaly detection and LLM root-cause analysis built in.

- Hybrid retrieval (vector + keyword) over ChromaDB with recency re-ranking, served by Gemini and exposed as REST, WebSocket, and an MCP server (9 tools).
- Detection ensemble — Isolation Forest (point) + LSTM Autoencoder (sequence) feeding a state-machine alert manager and Gemini incident reports.
- Custom simulator producing ~4.7M rows over 18 months; async FastAPI backend, React 19 console, human-in-the-loop by design.

Gemini · ChromaDB · MCP · PyTorch · scikit-learn · FastAPI · React 19

DataSense — Privacy-Preserving Automated EDA

PROTOTYPE

Upload a CSV, get a full statistical audit, a model recommendation, and plain-language insights — without sending raw data to any LLM.

- 7-step async pipeline — structural profiling, quality & correlation checks, Simpson's-paradox flags, PCA/clustering, and a confidence-scored model recommender.
- Privacy-aware LLM layer (Ollama local / OpenAI / Groq) — deterministic fact extraction so only computed facts, never raw rows, reach the model.

Paper accepted for oral presentation (to appear).

FastAPI · Next.js 14 · PostgreSQL · Polars · SciPy · Ollama

02 RESEARCH

forecasting · medical AI · reinforcement learning

ML-Enhanced SEIR Pipeline for Dengue Outbreak

Forecasting

Hybrid SEIR + ML for short-term dengue forecasting and spatial risk mapping across Sri Lanka.

[ONGOING]

Composite Clinical Proxy for Durable Benefit in

Oncology

Modeling durable clinical benefit from real-world clinicogenomic data.

[UNDER REVIEW]

Recovering Neuromodulatory Profiles from RL-Agent

Behaviour

Studying joint identifiability and degeneracy when inferring neuromodulatory profiles from behaviour.

[ONGOING]

Also: genetic algorithms applied to combinatorial optimization (TSP) and ML feature selection.